

Is Costa Rica's Transition Fast Enough for 1.5°C?

CO₂, GDP and Transport in a Supposed Green Champion

Group PG-20

Plan

1. Q1 – Are we on a 1.5°C path?

If we assume a simple linear decline in CO₂ per capita from 2010 to net-zero by 2050, how far is Costa Rica above or below that pathway today?

2. Q2 – Decoupling compared with similar economies

Among countries that had similar GDP per capita to Costa Rica in 2010, has Costa Rica done unusually well (or not) at growing income while stabilising or reducing CO₂ per capita?

3. Q3 – Is transport now the main bottleneck?

Given that Costa Rica's electricity is already mostly low-carbon, to what extent are transport emissions now dominating its remaining CO₂ and limiting further progress?

Data & approach

- **Country focus:** Costa Rica
- **Core data:** CO₂ emissions, CO₂ per capita, GDP per capita, and sectoral emissions
- **Source:** Our World in Data – CO₂ and Energy
- **Period:** From 1960s to early 2020s, with a focus on 2010 onwards for the 1.5°C pathway

Q1 – 1.5°C pathway

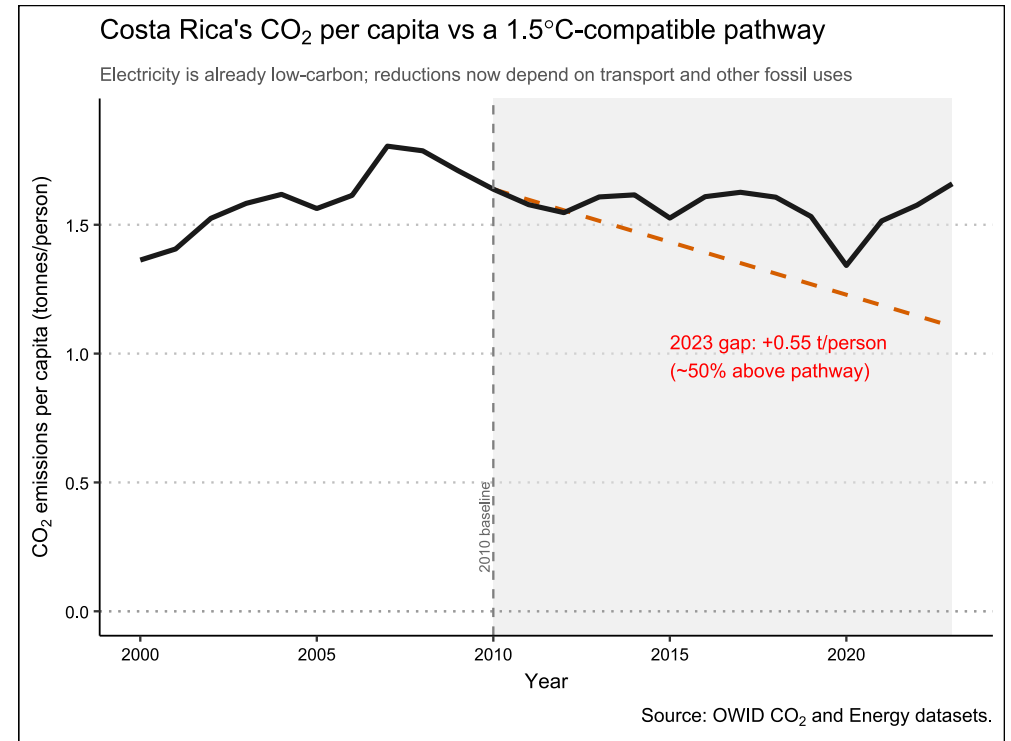
Compare observed CO₂ per capita with a **simple linear net-zero-by-2050 path** starting in 2010.

Q2 & Q3 – Structure and sectors

- Build a **peer group** of countries with similar GDP per capita in 2010
- Compare **GDP vs CO₂ per capita changes** and then zoom in on **transport vs other sectors** for Costa Rica

Q1 – Are we on a 1.5°C path?

- I use a **simple stylised pathway**: CO₂ per capita declines **linearly** from 2010 to **net-zero** in 2050.
- Costa Rica's **observed CO₂ per capita** is then plotted against this pathway.
- The **gap in the latest year** shows how far Costa Rica is **above** the 1.5°C-style path, under these assumptions.
- This is not a full carbon budget model – it's a **transparent, back-of-the-envelope benchmark**.



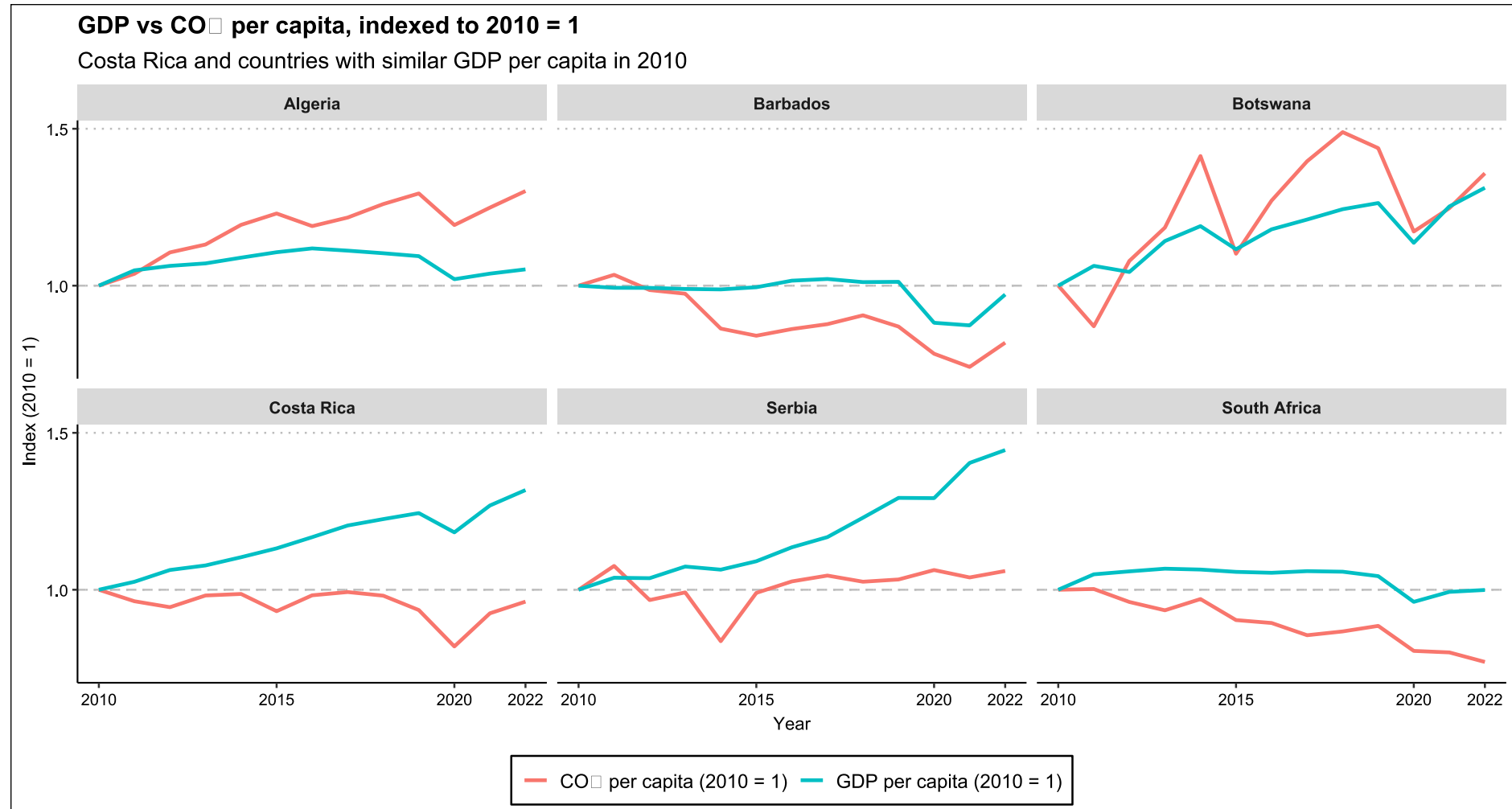
Q2 – GDP vs CO₂ change since 2010

- Here I summarise **total change from 2010 to the latest year** for each country.
- **GDP_pc_change**: growth in income per person.
- **CO₂_pc_change**: change in CO₂ per person.
- Costa Rica should stand out as **strong GDP growth with flat or slightly falling CO₂**, whereas some peers grow both.

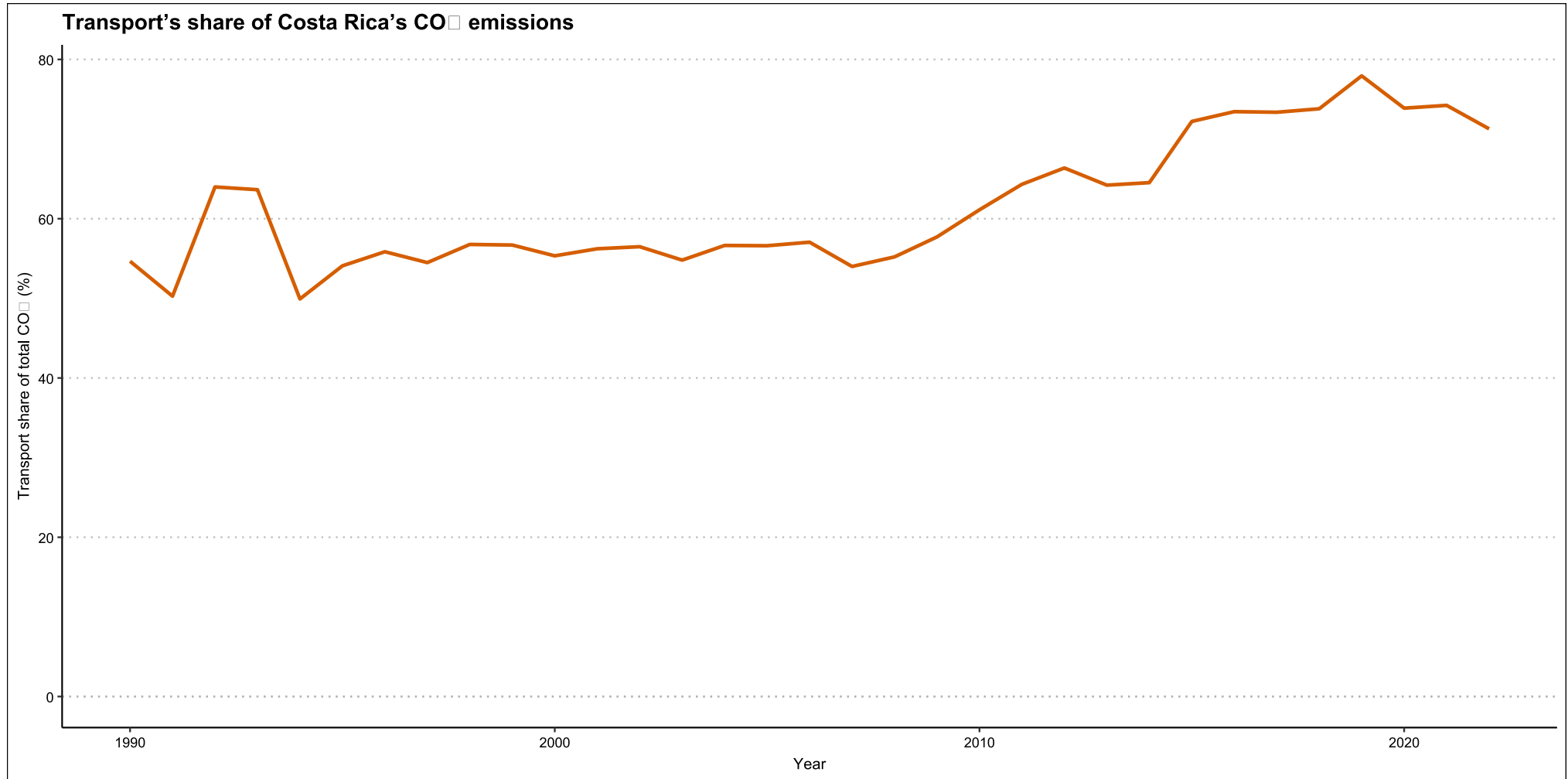
Change in GDP and CO₂ per capita since 2010

country	Start year	End year	GDP % change	CO2 % change
Serbia	2010	2022	+44.5%	+6.0%
Costa Rica	2010	2022	+31.8%	-3.8%
Botswana	2010	2022	+31.2%	+35.8%
Algeria	2010	2022	+5.2%	+30.2%
South Africa	2010	2022	-0.1%	-23.0%
Barbados	2010	2022	-2.8%	-18.2%

Q2 – Decoupling compared with similar economies



Q3 – Has transport become the main bottleneck?



Q3 – CO₂ per capita: transport vs other sectors

